1. Team Purpose or Mission

What does your team hope to accomplish by the end of the project? What questions will your dashboard help users explore, and who is the intended audience? How will your combined skills and interests support the work? What can your audience expect in terms of analytical quality, clarity, and professionalism?

By the end of the project, we hope to complete analyses that answer questions including, but not limited to:
1. What role do wetlands play in wildfire movement across the landscape?
2. Are wildfires less frequent near wetlands?
3. Has wetland loss occurred due to wildfires?
4. Do fires cross wetland boundaries?
Intended Audience: Wildfire and wetland researchers

Kat and Mason are taking Wetlands Ecology and can contribute relevant knowledge of wetland ecosystems. They are also taking GIS courses and can use ArcGIS to complement our Python-based analyses through spatial analysis and visualization. Sophia has experience working with and interpreting data, which will help generate analytical ideas and identify appropriate statistical methods. Together, we will deliver high-quality analyses supported by clear, well-written interpretations, while transparently acknowledging the limitations of our methods and findings.

2. Communication Plan and Collaborative Coding

- Communication (urgent questions): whatsapp, respond within 1 working day; Not expected to respond during weekends/late nights
- Sharing files/code: Github
- When are members generally available to work:
    - Sophia: Monday/Wednesday/Friday
    - Kat: Mon/Tues/Wed Afternoons, Friday
    - Mason: Monday/Wednesday/Friday
- When will you meet regularly:
    - Class time - Thursday 1:25-4:00 pm
    - If need extra time: Determined week of as needed
- How will you document decisions and next steps: Keep a decision/next step log in the documents file

3. Roles and Responsibilities

- What interests each member about the project?
    - Wetlands - Do wetlands serve as a natural barrier when wildfires happen? Are wetlands being burned? Are there fewer fires near wetlands?

    - Sophia: Are wetlands more effective in stopping fast or slow wildfires? What if we have more/less wetlands?
    - Mason: I am curious about wetlands generally speaking and their role in the environment
    - Kat: I am interested in learning about how wildfires and wetlands interact in California, and if there are fewer fires near wetlands.
- What strengths do you bring, and what skills do you want to develop?
    - Sophia: Python skills/analysis interpretation; Want to learn more about spatial data.
    - Kat: ArcGIS/spatial analysis skills, currently taking Wetlands Ecology. Want to learn more about spatial analysis and making map graphics in Python.
    - Mason: Also in Wetlands Ecology and Management, ArcGIS and R skills for geospatial analysis. I want to develop skills working with geospatial data in python.

How will you divide responsibilities for finding and preparing data, exploratory analysis, dashboard development, interpretation, documentation, and presentation? How will you ensure that everyone contributes meaningfully and understands the complete project? Identify who will review each other's work and how responsibilities may change as the project develops.

Wk	Milestones	Internal Milestones	Responsible for Milestone	Deadline
0	Defining research questions and project scope	- Define question - Pick geographic scale (CA?) - Define timescale (what years)	ALL	10/20
1	Assessing data feasibility	- Locate and upload datasets and deal with size constraints - Fire perimeters - Wetlands - Finalize team charter/submit work	Kat - Fire dataset Mason - Wetlands dataset Sophia - Team charter	10/20
2	Preparing and combining datasets	- Clean datasets - Subset by year - merge	TBD after 10/22 class time	10/27
3	Conducting exploratory analysis	- Examine fire locations/frequencies/etc	TBD after 10/22 class time	11/3

		- Examine wetlands locations/frequencies/etc - Look at overlaps, occurrences, etc		
4	Building a dashboard prototype	- Design “story” workflow - Finalize visualizations - Draw conclusions based on analysis	TBD after 10/22 class time	11/10
5	Reviewing and testing the dashboard/analysis	- Create presentation from dashboard - Continue to review and edit analysis/conclusions	TBD after 10/22 class time	11/17
6	Preparing the final presentation	- Complete the final reproduction check - Rehearse the presentation - Verify all submission links and access permissions.	TBD after 10/22 class time	11/19

4. Operating Norms or Ground Rules

How will you prepare for and run meetings? How will you assign tasks, set internal deadlines, and track progress? What should teammates expect when someone completes work independently—for example, organized files, readable code, and documented data-cleaning decisions? What will you do if someone anticipates missing a meeting or deadline? How will you combine and review work before submitting it?

- Prepare for meetings by doing work discussed ahead of time and communicate ahead of time if an adjustment is needed.
- Tasks generally organized by the team charter, but specified and adjusted the week before. If someone anticipates missing a meeting or deadline, responsibilities can be taken by other members or deadlines can be adjusted depending on current needs.
- General updates are seen in GitHub file updates, also notify group Whatsapp.
- All group members will review each milestone before turning in.

5. Values and Behaviors

What does being a good teammate mean to your group? Agree on expectations for reliability, respectful communication, constructive feedback, and openness to different ideas. How will you create space for all members to contribute and ask for help? How will you address uneven participation or concerns about the quality of someone's work?

- Everyone tries their best to contribute equally to the project and follow set deadlines, with clear communication if it's a busy week or you are behind on deadlines
- Ask for help when you need it, respond calmly to teammate concerns
- If bringing up concerns, have a kind and honest conversation

6. Resources and Constraints

What resources can support your work, including course materials, dataset documentation, instructors, TAs, and relevant subject-matter sources? What constraints should shape your scope—for example, the six-week timeline, missing data, differences in geographic or temporal coverage, technical skills, or other commitments?
Since follow-up with project partners may not be available, how will you resolve unanswered questions, document assumptions, and communicate limitations in the dashboard?

- Available resources
    - 710 course materials, labs, and homeworks
    - Library data visualization center
    - Instructors and classmates
    - Online resources
- Constraints
    - Data availability, size, and format
    - 6 week timeline in a busy part of the semester
    - Working with new and developing Python skills
- Resources and constraints will be documented in a file in the project for the audience to read. Contact emails can also be provided in the dashboard for follow-up questions.

7. Decision-Making Guidelines

How will you make decisions about research questions, datasets, analytical methods, and dashboard design? How will you ensure that relevant teammates are involved and that choices are supported by evidence? What will you do if you cannot reach consensus? When should you seek guidance from an instructor or TA?

We will make decisions collaboratively, considering each member's expertise, project goals, data availability, and feasibility. Choices will be supported by evidence from relevant literature, data exploration, and analytical results. If we cannot reach consensus, we will discuss trade-offs and prioritize the approach best aligned with our research objectives. We will seek guidance from the instructor or TA when we reach a milestone or when we encounter unresolved disagreements.

8. Goals and Objectives

What would a successful project look like? Identify a small set of concrete goals and how you will assess them. Consider whether the dashboard answers your research questions, uses appropriately prepared data, presents accurate and interpretable analyses, and is accessible and easy to navigate. What checks will you use to evaluate data quality, dashboard functionality, and the clarity of your findings? Distinguish essential deliverables from optional features.

A successful project will address our research questions while generating meaningful insights that deepen both our understanding and the audience's knowledge of the relationship between wetlands and wildfires.

Research Questions	Successful Project Will Have
1. What role do wetlands play in wildfire movement across the landscape?	Data-driven analysis of the relationship between wetlands and wildfire occurrence and spread, identifying potential patterns in how wetlands influence fire behavior.
2. Are wildfires less frequent near wetlands?	Quantitative comparison of wildfire frequency within a defined distance of wetlands versus areas farther away, using specified distance thresholds or buffer zones.
3. Has wetland loss occurred due to wildfires?	Assessment of changes in wetland area before and after nearby wildfire events to identify potential wetland loss associated with wildfire occurrence.
4. Do fires cross wetland boundaries?	Spatial overlay of wildfire perimeters and wetland boundaries to visualize and quantify the extent to which wildfires intersect with or extend across wetlands.

Ways to check-

Data quality: Check for missing values, duplicates, inconsistencies, and potential errors in the datasets. Verify that data sources, definitions, geographic boundaries, and time periods are consistent and appropriate for our research questions.

Dashboard functionality: Ask classmates who are not part of our team to test the dashboard and provide feedback on its ease of navigation, clarity, and usefulness. Evaluate whether users can easily find relevant information and answer the research questions using the dashboard.

Clarity of findings: Appropriate statistical tests to assess the robustness and significance of our findings. Ensure that visualizations and written interpretations communicate the results clearly, accurately reflect the analyses, and acknowledge relevant limitations and uncertainties.

Optional:

1. Comparing against other literatures
2. Dashboard form controls

9. Schedule and Milestones

Work backward from the assignment deadlines to establish internal milestones for:

- Defining research questions and project scope.
- Assessing data feasibility.
- Preparing and combining datasets.
- Conducting exploratory analysis.
- Building a dashboard prototype.
- Reviewing and testing the dashboard.
- Preparing the final presentation.

Account for each member's other commitments and leave time for revisions. Who is responsible for each milestone, how will you track progress, and when will you revisit the scope if necessary?

- **See roles and responsibilities section above.**